Identity as a Conserved Quantity: A Signal-Theoretic and Information-Theoretic Approach to the Ship of Theseus Paradox

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***Abstract*—The Ship of Theseus paradox has traditionally been viewed as a philosophical problem concerning persistence, continuity, and identity.**

This paper proposes a preliminary mathematical framework in which identity is modeled as a conserved quantity of a dynamical system rather than a property of its physical substrate.

Drawing inspiration from signal processing, information theory, graph theory, and dynamical systems, we introduce the concept of an Identity Functional capable of remaining invariant under admissible transformations.

The proposed framework is intended as a research program rather than a completed theory. Potential applications include cognitive systems, artificial intelligence, mind uploading, digital twins, teletransportation thought experiments, and semantic information preservation.

I. Introduction

The Ship of Theseus is one of the oldest and most influential paradoxes concerning identity.

Suppose a ship undergoes gradual replacement of all its wooden components over time.

After every plank has been replaced, an important question emerges:

> Is the resulting ship still the same ship?

Traditional approaches generally focus on:

- material continuity,
- physical persistence,
- psychological continuity,
- causal continuity.

This work explores an alternative possibility:

> Identity may be better understood as a conserved structural property rather than a material property.

II. Motivation from Engineering Systems

Modern engineering systems routinely preserve functionality despite complete hardware replacement.

Examples include:

- software migration,
- cloud infrastructure,
- signal reconstruction,
- distributed databases,
- neural network deployment across hardware platforms.

In these situations, physical realization changes while system-level behavior may remain unchanged.

This motivates the question:

> What quantity remains preserved when a system is considered the same despite physical transformation?

III. The Ship of Theseus as a Dynamical System

Let

$$S(t)$$

represent the state of a system at time t.

The replacement process may be represented as

$$S_0 \to S_1 \to S_2 \to \cdots \to S_n$$

where each state contains progressively different physical components.

The central question becomes:

$$S_n \stackrel{?}{=} S_0 \tag{1}$$

with respect to identity.

IV. Identity Functional

We introduce an Identity Functional

$$I(S)$$

defined over a system state.

Identity is preserved if

$$I(S_t) = I(S_0) \tag{2}$$

for all admissible transformations.

The functional is intended to capture structural, informational, or semantic properties of the system rather than its material composition.

V. Identity Conservation Principle

Inspired by conservation laws in physics, we propose the following research hypothesis:

> There exists an identity functional whose value remains invariant under identity-preserving transformations.

Mathematically,

$$\frac{dI}{dt} = 0 \tag{3}$$

under admissible transformations.

At present, the explicit form of I remains unknown.

Determining its mathematical structure is the primary objective of the proposed research program.

VI. Signal-Theoretic Interpretation

Consider a discrete signal

$$x[n]$$

and a transformed signal

$$y[n].$$

The similarity between both signals may be measured through normalized cross-correlation:

$$\rho(x, y) = \frac{\sum_n x[n]y[n]}{\sqrt{\sum_n x[n]^2}\sqrt{\sum_n y[n]^2}} \tag{4}$$

When

$$\rho(x, y) = 1 \tag{5}$$

the signal structure is perfectly preserved.

This suggests an analogy:

physical substrate may change while structural identity remains invariant.

VII. Information-Theoretic Interpretation

Let

$$H(S)$$

represent the information content of a system.

Identity preservation may require:

$$H(S_t) = H(S_0) \tag{6}$$

or more generally

$$D(S_t, S_0) \to 0 \tag{7}$$

where D denotes an information distance metric.

Possible candidates include:

- Mutual Information,
- Kullback-Leibler Divergence,
- Jensen-Shannon Distance,
- Kolmogorov Complexity.

VIII. Graph-Theoretic Interpretation

Let

$$G = (V, E)$$

represent a graph describing a system.

Nodes may represent:

- memories,
- concepts,
- components,
- functional units.

Edges represent relationships among them.

Identity preservation could then be defined through graph invariants.

Examples include:

- connectivity,
- spectral properties,
- homology groups,
- graph entropy.

A candidate identity metric becomes

$$I(G) = f(\lambda_1, \lambda_2, \ldots, \lambda_n) \tag{8}$$

where

$$\lambda_i$$

are graph spectral eigenvalues.

IX. Semantic Identity

An alternative interpretation focuses on meaning rather than physical representation.

Define

$$\Sigma(S)$$

as a semantic projection operator.

Two systems are semantically equivalent if

$$\Sigma(S_1) = \Sigma(S_2) \tag{9}$$

even when

$$S_1 \neq S_2. \tag{10}$$

This interpretation connects identity preservation to semantic invariance rather than physical continuity.

X. Potential Applications

Potential applications include:

- Artificial Intelligence,
- Cognitive Digital Twins,
- Mind Uploading,
- Teletransportation Thought Experiments,
- Human-AI Identity Transfer,
- Autonomous Agent Migration.

XI. Open Questions

Several unresolved questions remain:

1) Does a mathematically rigorous identity functional exist?
2) Can identity be measured experimentally?
3) Is identity fundamentally informational?
4) Is semantic preservation sufficient for identity preservation?
5) Are there observable invariants that survive complete substrate replacement?

XII. Conclusion

This work proposes a preliminary research framework in which identity is treated as a conserved quantity of a dynamical system.

Rather than focusing on material continuity, the framework investigates structural, informational, and semantic invariants as possible foundations of identity preservation.

The proposed approach transforms the Ship of Theseus from a purely philosophical paradox into a mathematical and computational research problem.

References

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